

Boundary Spanning Role of the IS Development Team in Consultant-Partnered Projects: Knowledge Management Perspective

Sanghoon Lee, Yonsei University, Korea

Bongsik Shin, San Diego State University, USA

Hogeun Lee, Yonsei University, Korea

ABSTRACT

This research examines the effect an Information Systems (IS) development team has on end-users' perceptions of system quality and system benefits by transferring relevant knowledge through inter- and intra-firm boundaries. The research context is the prevalent partnership in which an external consultant with relevant expertise leads a client team to undertake an IS project. A high-level research model that depicts dynamics among team competence (for knowledge transfer), consultant partnership (with the project team), knowledge transfer (through inter- and intra-firm boundaries), and project success is proposed grounded on the theory of boundary spanning. Key indicators of team competence and consultant partnership are derived from existing studies. With expected multiplicity in the conceptual dimensions of team competence and consultant partnership, they are designated as second-order constructs with first-order manifest variables. User perceptions of the post-implementation quality and benefits of an information system serve as project success variables. Relevant hypotheses propose dynamics among the studied constructs. Survey data are gathered from both system developers and end-users, and the integrity of the research model and corresponding hypotheses are empirically tested with structural equation modeling. Data analysis confirmed the importance of knowledge transfer for the post-implementation success of an IS project.

Keywords: Boundary Spanning, IS Success, Knowledge Management, Knowledge Transfer, Project Management

1. INTRODUCTION

There is a growing body of literature underscoring the importance of knowledge-driven business operations to create new business values

and opportunities (Alavi & Leidner, 2001; Karlsen & Gottschalk, 2003). Not surprisingly, knowledge resources and their management are being accepted as key predictors of organizational and team performance (Argote & Ingram, 2000; Tanriverdi, 2005; Jih et al., 2006). Recognizing the importance of intellectual assets as

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a competitive weapon, firms are putting more efforts into the effective creation, acquisition, sharing, and utilization of knowledge (Argote & Ingram, 2000; Nonaka, 1994).

Information systems (IS) projects within a firm offer an excellent opportunity for learning (Stein & Vandenbosch, 1996). In fact, knowledge acquisition itself has been suggested as the essential element of a successful project (Ayas, 1996; Edberg & Olfman, 2001; Schindler & Eppler, 2003). Through an IS project, a firm's employees can learn methods of process innovation and re-engineering, institutionalize best practices for systems development and deployment, and gain post-implementation know-how of system maintenance and management (Loo, 2003; Schindler & Eppler, 2003). The transfer of such knowledge takes place at different levels of interactivity (i.e., between developers and external partners, between developers and end-users, and among developers) (Bressman et al., 1999; Karlsen & Gottschalk, 2004). It has been asserted that successful knowledge management in an IS project can enhance a firm's core competence and performance (Eppler et al., 1996; Slaughter & Kirsch, 2006). Nonetheless, it is also true that many organizations fail to capitalize on IS projects in enhancing organizational intelligence (Schindler & Eppler, 2003; Stein & Valdenbosch, 1996).

The focus of our empirical research is to examine the effect an IS development team has on end-users' perception of system quality and system benefits by transferring relevant knowledge across inter- and intra-firm boundaries. The research context is the prevalent partnership in which an external consultant with relevant expertise leads the client team of a firm to undertake an IS project. For the study, a high-level research model that depicts dynamics among *team competence*, *consultant partnership*, *knowledge transfer*, and *project success* is proposed grounded on the theory of boundary spanning. Key indicators of team competence (absorptive capacity, project manager's leadership, team climate, and training) and consultant partnership (team trust, effectiveness

of communication, and levels of conflict) are derived from existing studies. With expected multiplicity in their conceptual dimensions, team competence and consultant partnership are designated as second-order constructs with first-order manifest variables. User perceptions of the quality and benefits of an IS are adopted as success variables.

Knowledge transfer, here, represents various knowledge spanning activities and processes (both structured and unstructured) across recognized boundaries at a firm. It therefore implicates acquiring, codifying, translating, amplifying, institutionalizing, and sharing both tacit and explicit knowledge (Kim et al., 2006; Alavi & Leidner, 2001; Nonaka, 1994). Given the mode of partnership in this study, knowledge transfer may be largely uni-directional from consultants to developers and from developers to organizational stakeholders including system users. For the bridge role, a project team effectively acts as gatekeeper, scout, ambassador, and guard of associated knowledge (Levina & Vaast, 2005). Also, it should be adept in creating and marketing various boundary objects (i.e., documents or manual) of a target information system (Levina & Vaast, 2005; Pawlowski & Robey, 2004). Naturally, a team's competence in effectuating inter- and intra-firm knowledge transfer and its partnership quality with consultants are crucial for the ultimate success of an IS project.

The theory of boundary spanning is discussed as the conceptual foundation and a research model that reflects dynamics among implicated factors is derived. Relevant hypotheses are proposed on the relationship among the studied variables. Survey data are gathered separately from the developer and the end-user of an IS in a pair-wise manner and combined for analysis. The integrity of the research model and corresponding hypotheses are then empirically tested based on structural equation modeling. Data analysis confirmed the vital role of a project team in the boundary spanning of system knowledge for the post-implementation success of an IS project.

2. THEORY

The theory of boundary spanning primarily focuses on a team's (or a group of agents') outward (rather than inward) or across-fields activities and their accompanying effects on the team and its organization. A team is a task-oriented group formed for problem solving (Streufert & Nogami, 1992) and therefore its members share common objectives (Orasanu & Salas, 1993). Ancona and Caldwell (1992) show that a team's boundary spanning activities can be largely classified into 15 distinctive types, including mapping, molding, and filtering. Such boundary spanning may be driven in either community-like or market-like practices (Levina & Vaast, 2006). In the former, related practices are embodied through inter-personal (both vertical and horizontal) relationships; meanwhile, in the latter, practices are more objectified through wider adoption of boundary objects. Levina and Vaast (2006) defined objects, such as documents, procedures, and roles, as "symbolic representations of practice and relations produced through practice." Because of the variety of activities in which it must take part, a team that engages in boundary spanning must also act as a knowledge broker (Levina & Vaast, 2006).

Knowledge transfer constitutes an important boundary spanning practice for IS professionals at an organization (Pawlowski & Robey, 2004). Transfer of knowledge (both explicit and implicit) is a process in which an individual, a team, a group, or a functional unit is affected by the experience of others (Argote & Ingram, 2000). An IS project team should be a boundary spanner that provides communication linkages (Pawlowski & Robey, 2004) among people with divergent knowledge backgrounds. This is especially true because IS projects are inherently knowledge-intensive, and a project's performance is affected by the integration of specialized knowledge necessary for different roles and phases of a process (Patnayakuni et al., 2006). In addition, boundary spanning of IS knowledge among heterogeneous fields of practice may be conditioned by such factors

as the characteristics of knowledge transferred (i.e., causal ambiguity), of the source (i.e., trustworthy source), of the recipient (i.e., lack of absorptive capacity), and of the context (i.e., arduous relationship) (Szulanski, 1996).

A team as a boundary spanner should be competent in facilitating knowledge transfer. For this, a team and its leadership play various roles, including gatekeepers (i.e., knowledge mediation and channeling), scouts (i.e., knowledge introduction), ambassadors (i.e., lobbying for knowledge resources) and guards (i.e., monitoring knowledge requests) (Levina & Vaast, 2005). These roles may be consolidated into the roles of technical scout (to relieve knowledge deficits) and task coordinator (to clear operational uncertainties) (Mitchell, 2006). From the perspective of situation learning, boundary spanners must be knowledge brokers who link multiple communities by translating local knowledge and who offer boundary objects (i.e., systems, documents, and artifacts) to convey interpretive and flexible knowledge to different communities and fields (Bechky, 2003; Pawlowski & Robey, 2004). These various roles highlight the importance of project team competence in integrating external and internal knowledge within and across organizational boundaries (Mitchell, 2006).

Traditional communication channels such as personal contacts, conferences, and training are instrumental for the flow of both tacit and explicit knowledge (Mitchell, 2006; Nonaka, 1994). Knowledge transfer can also be significantly effectuated through the mobilization of boundary objects including documents, reporting forms, manuals, information portals, prototypes, drawings, and IT applications (Levina, 2005; Levina & Vaast, 2005; Pawlowski & Robey, 2004). Boundary objects become effective when they are tangible, concrete (addressing specific problems), accessible, and up-to-date; benefiting local needs, overcoming constraints of implicated parties, and helping the formation of a collective identity (Levina, 2005). Slaughter and Kirsch (2006) relate the effectiveness to the intensity (i.e., frequency) and composition (i.e., types) of knowledge

transfer mechanisms. It is also noted that the quality and level of knowledge transfer is conditioned by knowledge-related (i.e., absorptive capacity), communication-related (i.e., source credibility), and motivational (i.e., intrinsic and extrinsic motivations) factors (Ko et al., 2005). Argote et al. (2000) discussed the conditions of knowledge transfer in terms of characteristics of the relationship among organizations, of knowledge transferred, of the organization, and of the transfer process.

The effect knowledge transfer on system and organizational performance has been discussed. Methods and procedures for knowledge delegation affect project success, including client benefits and system performance (Awazu, 2006; Karlsen & Gottschalk, 2004). Ayas (1996) emphasizes that the success of a project over time is largely dependent on the capacity of its participants to learn from experience. Stein and Vandenbosch (1996) posit a comprehensive association between knowledge transfer and project success, and between project success and organizational performance. Loo (2003) links project outcomes with the technical and non-technical competence of the people involved. To realize this, the IS planning, implementation, and usage may be closely evaluated from the perspective of key stakeholders in different project stages (Adelakun & Jennex, 2002). It is also indicated that the portfolio of knowledge transfer mechanisms affect project performance factors such as on-time project completion (Slaughter & Kirsch, 2006). Patnayakuni et al. (2006) reveal the significant role of process formalization in the relationship between knowledge integration capacity and project outcomes. Importance of knowledge transfer in sustaining competitive advantage and in addressing business problems is discussed (Jasimuddin, 2007; Martin et al., 2008).

Argote and Ingram (2000) explain the association between knowledge and project success in terms of fit among three knowledge embodiments: members (human components), tools (software and hardware), and tasks (organizational goals and objectives). According to them, much of tacit knowledge is embed-

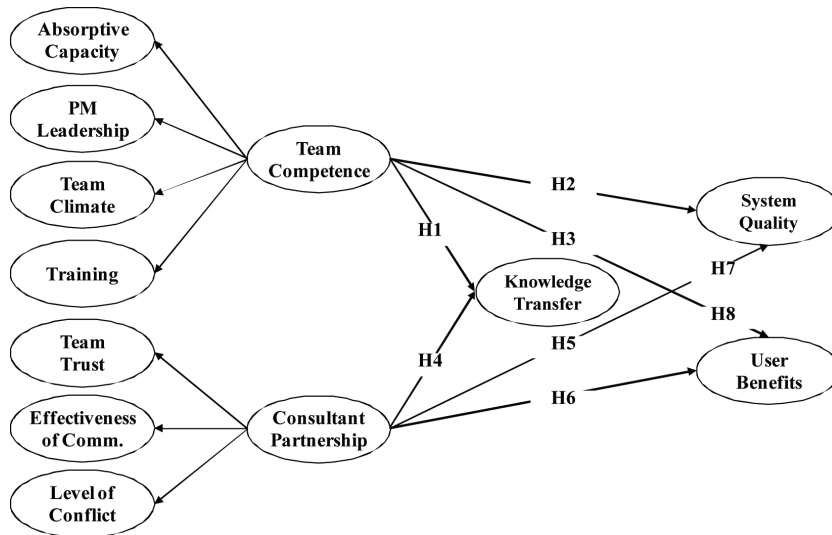
ded in members, and more explicit, codifiable knowledge is embedded in tools. These three basic knowledge elements are combined to form subnetworks such as member-member (i.e., social network), tool-tool (i.e., combination of technologies used by the organization), and task-task (i.e., sequence of tasks or routines an organization uses) networks. The success of an IS project may be assessed in terms of internal fit within a sub-network (i.e., member-member) and external fit between subnetworks (i.e., member-tool) (Argote and Ingram 2000). Therefore, in the context of the consultant-driven IS project, partnership quality between the external consultant and the client team is an important member-member subnetwork that influences subsequent project success.

Given the expected role of team competence and consultant partnership, knowledge-disseminating practices implemented by the developer team across inter- and intra-firm boundaries may prompt a better external fit between members (i.e., end-users) and tools (i.e., system), and between tools and tasks. The member-tool fit represents the readiness of user knowledge to take advantage of a system. A team's effort toward knowledge delegation across intra-firm boundaries should improve member-system fit, ultimately leading to enhanced system benefits among end-users. For this, a project team and its leadership should act as knowledge brokers and integrators who can import external knowledge and transform it into a form to be consumed by internal clients (Pawlowski & Robey, 2004). The level of tool-task fit becomes an indicator of system quality in supporting target tasks. A team's efforts toward knowledge acquisition and internalization of system knowledge across inter-firm boundaries are expected to have a positive effect on the quality of a target system, and subsequently the perception of system quality among end-users.

3. RESEARCH MODEL

The theoretical discussion above underscores the important role of two constructs, team com-

Figure 1. Research model



petence in knowledge transfer and the quality of consultant partnership, in disseminating system-knowledge and ensuring the post-implementation success of an IS project. The two constructs are closely associated because a team's capability in accessing external knowledge and assimilating it internally (Freeze & Kulkarni, 2008; Mitchell, 2006) must be optimized in effective collaboration with external consultants (member-to-member fit). Therefore, a research model (Figure 1) is derived to depict the role of team competence and consultant partnership in promoting effective knowledge transfer and in assuring the post-implementation success of an IS project. Due to the multiplicity of conceptual dimensions for team competence and consultant partnership, we chose an approach in which they become second-order constructs.

A significant body of research indicates that the knowledge transfer capacity of a development team is affected by its individual- and group-level characteristics. For example, Szulanski (1996, 2000) emphasizes the importance of such characteristics as leadership, motivation, trust, and absorptive capacity for effective knowledge transfer. Baldwin and Ford (1988) highlight the role of personal ability and train-

ing elements. Cultural implications are frequently examined as well (Bock et al., 2005). Drawing from extant studies, four variables that could be particularly discernible in manifesting the level of team competence are identified: absorptive capacity, project manager (PM) leadership, team climate, and level of training. The variables are briefly defined here with a more detailed discussion offered during the hypothesis development.

1. Absorptive capacity represents a team's ability to acquire and process knowledge. Absorptive capacity of a team may not necessarily be the simple summation of each member's ability. Nonetheless, individual ability and the level of individual readiness to absorb and digest knowledge are fundamental for group-level processing and subsequent organization-wide infusion (Cohen & Levinthal, 1990).
2. Project manager (PM) leadership can either augment or degrade team performance (Hackman, 1987; Tannenbaum et al., 1992). In a goal-driven organization, leadership affects not only individual behaviors and attitudes, but also the dynamics of group

activities (Bass 1990). It is, therefore, reasoned that PM leadership is an important indicator of team competence for knowledge transfer practices.

3. Team climate represents socio-psychological attitudes shared among members toward decision-making, task undertaking, and reward structures (Ashforth 1985; Joyce and Slocum 1982). It is a form of social culture that may promote or inhibit team members' tendency toward knowledge sharing (Usoro & Kuofie, 2006; Karlsen & Gottschalk, 2004).
4. Level of training: Training is an essential component to increase competence of a team or an individual (Rockart & Flannery, 1983). Naturally, it is an effective means to achieve goals associated with an IS project. The motivation and readiness of a team in spanning the boundaries and disseminating IS knowledge may naturally be conducive to the level of technical and non-technical training offered (Szulanski, 1996).

Consultant partnership, another second-order construct, represents the quality of relationship between the consultant and the developer team. The partnership is a purposive strategic relationship where interdependent partners share compatible goals and strive for mutual benefits (Mohr & Speckman, 1994). Trust, communication, and conflict are considered as defining elements in judging the partnership of an IT project (Henderson, 1990; Lee & Kim, 1999). Therefore, we identified three manifest variables of consultant partnership: team trust in consulting partners, effectiveness of communication, and the level of conflict.

1. Team trust in consulting partners is the unwritten expectation that partners will behave in a predictable and acceptable manner (Sako, 1991). It carries conceptual similitude to source credibility (Ko et al., 2005) that a team has with regards to a knowledge source. Trust determines the

quality of partnership (Henderson, 1990) and enhances collaboration among people working on a target project (Herzog, 2001). The positive effect of trust also applies to knowledge management (Ryoo and Kim, 2009). Stein and Vanderbosch (1996) suggested that the developers' view of (i.e., confidence in) an expert brought into an organization is associated with their learning.

2. Effectiveness of communication: Because of the initial knowledge gap for a target system, teams and consultants should engage in effective communication in order to delegate project knowledge (Ko et al., 2005). Consultants' communication capability may be especially vital in strengthening the partnership because they are the providers of advanced, and frequently unstructured and tacit, know-how. Also, better communication increases intimacy, lowering emotional barriers (Szulanski, 1996) and effectuating boundary spanning of system knowledge.
3. Level of conflict: Conflict is the manifestation of difficulties in a relationship. Conflicts between consultants and developers may be partially attributable to a lack of shared understanding in terms of work values, philosophy, problem-solving approaches, and prior work experience (Ko et al., 2005). The relationship between knowledge sources and knowledge recipients affects the level of knowledge transfer (Argote & Ingram, 2000). When the relationship is more arduous than cooperative, it may negatively affect the partnership between two parties and adversely influence knowledge transfer (Ko et al., 2005).

In our study, user perceptions of system quality and user benefits are adopted as the success variables of an IS project. Project success can be judged qualitatively and quantitatively (Adelakun & Jennex, 2002). Baccarini (1999) and Mitchell (2006) classified success variables

in terms of process or project management-related and product-effects related. The former pays attention to the procedural quality of a project, including such factors as cost effectiveness, time management, and the accommodation of stakeholders' inputs. The latter relates to the delivered system as a product and its subsequent effects. These include system quality, user benefits, strategic contribution to a business goal, improvements in target processes, user satisfaction, and organizational learning (DeLone & McLean, 1992). *System quality* and *user benefits* are two requisite variables of product effects and adopted in our study as indicators of project success. System quality in this study represents the quality of IS implementation perceived by users. Slightly differ from the DeLone and McLean's (1992) definition of system quality that emphasizes the system aspects per se, we use the term as the manifestation of system acceptance, usage, and satisfaction among end-users. User benefits measure changes a system incurred to end-users in terms of their productivity, efficiency in decision-making, and learning (Lee et al., 2009). Although both are project success variables (DeLone & McLean, 1992), these variables may not necessarily covary because of their conceptual divergence.

Figure 1 shows the research model that includes the first- and second-order latent variables drawn from our theoretical discussion and extant studies. In the model, project success is a direct function of the independent constructs. Additionally, consultant partnership and team competence are expected to affect the delegation of system knowledge among project stakeholders, especially consultants, project team members, and end-users. With the reflective (rather than formative) approach in defining the relationship between the first- and second-order constructs, we are taking the position that the first-order constructs are manifest variables of the second-order constructs. Also, with the reflective model, we presume the existence of covariation among the first-order constructs

and the second-order constructs explain the covariation (Chin, 1998).

4. HYPOTHESES DEVELOPMENT

4.1. Team Competence, Knowledge Transfer, and Project Success

The team competence variable represents a team's capability of spanning system knowledge through inter- and intra-firm boundaries. It is a team's ability to acquire external knowledge from consultants and to internalize (or synthesize) it (Mitchell, 2006). Team competence may be a function of a team's capacity for effective collaboration and coordination across organizational and professional boundaries (Levina, 2005). Team competence may be more critical in sharing tacit rather than explicit knowledge (Nonaka, 1994).

Several studies indicate that the transfer of system knowledge among stakeholders has a positive influence on project success. Patnayakuni et al. (2006) point out that the integration of explicit and implicit knowledge across different phases of system development increases project performance. Mitchell (2006) discusses the effect of knowledge deficit in terms of performance risk (outcomes are assessed inaccurately) and execution risk (outcomes are unsatisfactory) of an IS project. Slaughter and Kirsch (2006) understand the performance effect in terms of the composition and intensity of knowledge transfer mechanisms. The associative relationship among three constructs (team competence, knowledge transfer, and project success) is also implied by studies that examine the particular role of the first-order constructs in knowledge transfer.

Absorptive capacity: An IS project with external partners is an opportunity through which know-how such as best practices is obtained, transformed and diffused internally. A team's absorptive capacity

is largely determined by the members' individual readiness to absorb necessary knowledge (Cohen & Levinthal, 1990). That is, even when knowledge is given to a team, it may not be absorbed if team members lack the necessary competence (O'Dell & Grayson, 1998). There may be especially relevant attributes (i.e., IS expertise and project experience) that affect individual preparedness, and therefore the absorptive capacity, of a team (Cohen & Levinthal, 1990). A firm's absorptive ability influences its effectiveness in terms of organization-wide education and management (Lane et al., 2001). This implies that a team's absorptive capacity is an important precursor of its dynamic and combinative capability (Mitchell, 2006), which can enhance the team's knowledge integration, and, ultimately, the performance of an IS project.

Project manager (PM) leadership: The PM's key role in boosting team competence is indisputable. The PM uses proper leadership techniques to set up general as well as detailed goals and objectives, determines necessary project rules and guidelines, specifies the roles and activities of participating members, and manages and allocates available resources. One requisite element of project resources is knowledge. APM can encourage boundary spanning of IS knowledge by setting goals and objectives of knowledge delegation, cultivating collaborative teamwork, utilizing strengths of team members, preventing possible conflicts, and introducing motivational measures to promote knowledge sharing. By accessing external knowledge (or an "outside view"), a PM can also avoid cognitive biases during a project (Mitchell, 2006). It is, therefore, reasoned that PM leadership plays an important role in effectuating knowledge transfer during the course of an IS project (Szulanski, 1996). Occupying a leadership role in knowledge spanning, the PM strongly influences team performance and therefore has been

frequently regarded as key to the success of an IS project (Jiang et al., 1998; Lee & Sweeney, 2001; Tannenbaum et al., 1992).

Team climate: The role of social cultures such as positive team climate as a promoter or inhibitor of knowledge transfer has been studied (Karlsen & Gottschalk, 2004). Knowledge sharing may be highly conducive to positive group dynamics that may manifest themselves in various forms. A positive team climate fosters innovation and constructive changes, member coherence and a cooperative spirit, respect for self-control and shared decision-making, trust among team members, and healthy tension among associates (Amabile, 1988; Angle, 1989; Bock et al., 2005; Pelz & Andrews, 2005). This implies that positive dynamics among team members is an indispensable element of team competence for effective knowledge transfer. A constructive team climate can allow team members to make the most of their abilities and augment intra- and inter-firm knowledge access and transfer. Such a healthy climate also affects the performance of a team project by encouraging member communication and participation (Amabile, 1988; Pelz & Andrews, 1966; Roffe, 1999).

Level of training: Training is an effective means to ensure the success of a project: to attract prospective users, improve service quality, increase adoption and usage, and increase user satisfaction with a system (Nelson & Cheney, 1987; Sanders & Courtney, 1985). The motivation and readiness of a team to span knowledge boundaries of an IS project is naturally related to the level of their technical and non-technical training. Baldwin and Ford's (1988) study, for example, suggests that training influences the effectiveness of knowledge acquisition. Adequate training in both technical (i.e., development methods) and managerial (i.e., strategic implications) aspects of an IS may enable team members to better understand the

importance of institutionalizing acquired knowledge (Szulanski, 1996). In addition, the importance of training in reinforcing team competence and as a critical success factor of an IS project has been repeatedly emphasized (Barquin & Edelstein, 1997; Bruwer, 1984; Goodhue & Thompson, 1995; Nelson & Cheney, 1987).

Although each manifest variable itself can be treated as a direct antecedent of knowledge transfer and project performance, we posit their covariation as the first-order predictors of team competence for knowledge transfer. The discussion of manifest variables suggests that greater competence at the team level is not only an effective force in mediating system knowledge across external-to-internal and internal-to-internal boundaries, but also facilitates IS project success and its resulting benefits. We therefore offer the following hypotheses:

Hypothesis 1: The competence of an IS project team is positively associated with knowledge transfer through inter- and intra-firm boundaries.

Hypothesis 2: The competence of an IS project team is positively associated with the perception of system quality among end-users.

Hypothesis 3: The competence of an IS project team is positively associated with the perception of system benefits among end-users.

4.2. Consultant Partnership and Knowledge Transfer

Consulting partners' know-how about a particular system (e.g., ERP) and their close partnership with the team is expected to result in a better inter-firm flow of system knowledge, therefore increasing the chances of project success (Szulanski, 1996). The collaborative, communicative and trusting partnership between consultants and project team members enhances knowledge transfer through various social and technological mechanisms (Slaughter

& Kirsch, 2006). The associative relationship between consultant partnership and knowledge transfer is also evidenced by existing studies that examine the effect of the first-order constructs (team trust in consulting partners, effectiveness of communication, and levels of conflict) on the quality of partnership and learning effects.

Team trust in consulting partners: It has been suggested that IS developers' confidence in an external expert can boost their learning effect (Stein & Vanderbosch, 1996). Similarly, studies indicate that member trust improves team learning because it promotes knowledge flow and reduces opportunistic behaviors (Lee & Kim, 1999; Szulanski, 2000). Consultants, for their part, can influence team members more effectively when the latter trust the former (Perloff, 1993). These studies give a strong indication that team trust in external consultants is important for effective dissemination of system knowledge. Studies also suggest that trust is an intangible asset for team performance (Henderson, 1990) and a key predictor of partnership success (Aulakh & Kotabe, 1996; Kale et al., 2000). Their association effect may be especially salient for large-scale IS projects characterized by complexity and uncertainties.

Effectiveness of communication: The importance of communication for a successful partnership cannot be overstated. Good communication is indispensable for knowledge transfer between knowledge bearers and knowledge recipients (Szulanski, 1996). Given the initial asymmetry of system knowledge and its subsequent flow between two parties, effective communication should curtail emotional barriers, promote a collaborative spirit, and convey relevant IS knowledge to receiving parties (Ko et al., 2005; Szulanski, 1996). It has also been shown that adequate communication among involved parties improves the performance of group tasks. For

instance, Keller (1994) showed that the degree of communication was the most important performance indicator in a software development project. It is, accordingly, expected that communication effectiveness between consultants and the team is a positive indicator of consultant partnership, improving knowledge transfer across the inter-team boundary and affecting the post-implementation success of an IS project.

Levels of conflict: Conflicts occur when people with different roles and responsibilities must cooperate to achieve a common goal (Tushman & Nadler, 1978). They are also attributable to differences in work values, philosophy, problem-solving approaches, and prior work experience (Ko et al., 2005). These conflicts may be viewed either as a negative force to be resolved (Ehtamo et al., 1999) or as a dialectic process to be adequately managed (Irish, 1975). Studies suggest that conflict resolutions may have more positive long-term effects when the involved parties, not a third party, are in charge of the mediating process (Assael, 1969) and when persuasion is the means of resolution, rather than an authoritative edict being enforced (Mohr & Spekman, 1994). Overall, the majority of studies regards conflict as a detrimental force and emphasizes its resolution or keeping it to a manageable level. Naturally, efforts or procedures to alleviate conflict may be necessary to sustain knowledge transfer and eventual project success. We take the position that the levels of conflict variable negatively covaries with the other two variables (communication effectiveness and team trust in consultants) in determining the level of consultant partnership. Based on the discussion above, it is hypothesized that:

Hypothesis 4: The quality of the partnership between the consultant and the project team is positively associated with knowl-

edge transfer across inter- and intra-firm boundaries.

Hypothesis 5: The quality of the partnership between the consultant and the project team is positively associated with the perception of system quality among end-users.

Hypothesis 6: The quality of the partnership between the consultant and the project team is positively associated with the perception of system benefits among end-users.

4.3. Knowledge Transfer and Project Success

IS projects led by consulting partners have become a valuable means of bridging knowledge gaps that exist between firms and within a firm. Knowledge of an IS includes rules, tools, guidelines, configuration, testing, and other best practices necessary for IS implementation, operation, maintenance, and usage (Ko et al., 2005). For example, many large-scale IS projects have been executed according to best practices packaged by consulting firms (Goodman & Darr, 1998). When best practices are highly standardized, projects can be relatively easy to execute. Many times, though, firms undertaking similar IS projects end up with opposite results. This is an indication that successful execution of standardized practices is a necessary, but not a sufficient condition for their immediate or long-term success. Among other success conditions, continuous improvement of the knowledge transfer structure within the project context may be especially requisite (Karlsen & Gottschalk, 2003; Mitchell, 2006; Wang et al., 2007). Knowledge transfer practices within a project team set the stage for introduction of knowledge into the hosting firm and its subsequent institutionalization (Edberg & Olfman, 2001). Effective assimilation of knowledge into the tangible or intangible memory structure of a firm can boost firm competence (Szulanski, 1996). The performance effect of knowledge sharing may be applicable to the firm's employees and internal departments.

The contribution effect of shared knowledge may be especially salient when IS knowledge, presented to a receiving individual, team, or firm, lacks standardization because of its tacit nature. Accordingly, the success of an IS project requires emphasis on learning that fortifies a firm's or team's ability to generate knowledge, make it explicit, and transfer it within and across team boundaries (Ayas 1996). We therefore propose that:

Hypothesis 7: Transfer of IS knowledge across inter-firm (i.e., consultant and project team) and intra-firm (i.e., project team and internal stakeholders) boundaries is positively associated with system quality as perceived by end-users.

Hypothesis 8: Transfer of IS knowledge across inter-firm (i.e., consultant and project team) and intra-firm (i.e., project team and internal stakeholders) boundaries is positively associated with system benefits as perceived by end-users.

5. RESEARCH METHOD

5.1. Survey Development

A survey questionnaire was used to gather data (Appendix A). The survey was developed in part (1) by borrowing validated items from current studies and (2) by converting the definitions of study variables into question items to fit the contexts of knowledge transfer and consultant partnership. Questions about absorptive capacity, IS implementation quality, and user benefits belong to (1) and the others belong to (2). The reference sources of survey items and their definitions are summarized in Table 1. Responses were measured on a 7-point Likert scale ranging from 1 (completely disagree) to 7 (completely agree). In order to avoid response bias, we divided the survey into two parts: the first part designed for a team member and the second part designed for an end-user. The first part (for team members) included question items about team competence, the consultant partner-

ship, and knowledge transfer. The second part (for end-users) contained items about project success measured in terms of perceived system quality and user benefits.

Three IT managers from three different firms and two consultants from two IT consulting firms reviewed the survey items; their content validity was improved based on the five reviewers' feedback. The reliability of survey items was further improved through a pilot test based on 39 survey responses. Data analysis indicated that, among the eight original items of the training variable, three had low factor loadings and were dropped from the survey. Survey responses used for the pilot testing were not included in the data analysis for hypothesis testing.

5.2. Data Gathering

We surveyed two people from each of the identified companies: a team member who participated in the development of a particular system and an end-user who utilizes the system to undertake organizational duties. Team members responded to question items about team competence, partnership quality, and knowledge transfer. End-users answered questions about system quality and user benefits. The combined set of responses from both parties became an observation point. The sampling frame was composed of 887 enterprises registered on the Korean stock market and additional 650 medium/small firms. The list of medium and small firms was provided by the institute that promotes the adoption of information systems. From the sampling frame, 560 firms were randomly selected, which included 250 publicly listed firms and 310 unlisted medium and small firms. The survey, with a cover letter that detailed the research goal and the intended use of survey data, was mailed to selected companies. Survey data were gathered through direct visits, email, and fax.

To ensure an adequate response rate, three follow-ups were made via email and telephone correspondence every two weeks. During the

Table 1. Sources of survey items

Constructs	Definition	Reference sources
Team competence: A team's capacity to facilitate inter- and intra-firm knowledge sharing		
Absorptive capacity (AC)	A team's ability to acquire and process knowledge	Suh and Kim (2002)
PM leadership (PL)	The effectiveness of PM leadership in project management	Weinkauff and Hogel (2002)
Team climate (TC)	Socio-psychological attitudes shared among project members	Abbey and Dickson (1983), Dailey (1978), Tushman and Nadler (1986), O'Reilly (1989), Quinn (1985), Jain and Triands (1990), VanGundy (1984)
Training (TR)	The degree of technical and non-technical training received	Zmud (1983), Lee, Trauth, and Farwell (1995)
Consultant partnership: The quality of partnership between the project team and the consulting partner		
Team trust (TT)	The degree of team trust in the consulting partner	Cho and Jeon (1998), Lee and Kim (1999)
Effectiveness of communication (EC)	Effectiveness of communication between the team and the consultant	
Level of conflict (LC)	The degree of conflict between the team and the consultant	
Knowledge Transfer (KT)	Effectiveness in transferring system knowledge through intra- and inter-firm boundaries	De Leo (1994), Kogut and Zander (1993), Marquardt (1996)
Project success		
System (SQ)	The degree of actual usage and user acceptance of a system	Karlsen and Gottschalk (2003)
User benefits (UB)	The degree of improvement in task efficiency and effectiveness	

ten-week period of survey gathering, 182 survey forms (each including both the team part and the end-user part) were returned, resulting in a 30 response rate. Then, 21 surveys were dropped due to their incompleteness, leaving 161 data points (from 161 companies with 322 responses) for analysis. Slightly more than half of the surveyed projects had been to develop ERP systems (53), followed by CRM (customer relationship management) (11) and KMS (knowledge management system) (9). The summary of responses clearly indicated that target systems implicated large-scale and complicated IS projects. Here, we would like to note that, although the survey was answered by individuals, responses to the variables of team competence, partnership quality, and knowledge

transfer were grounded on the respondent's view of team-level characteristics.

6. ANALYSIS AND RESULTS

We used two different tools for data analysis: SPSS 11.0 for the descriptive statistics and LISREL 8.50 to validate the integrity of the research model and proposed hypotheses.

6.1. Descriptive Statistics

ERP projects have been prevalent since the mid-90s, and the survey reflects this fact. The descriptive statistics confirm that more than 50% of the enterprise IT projects was ERP-related that is frequently regarded as the most convo-

Table 2. Descriptive statistics

Project	Freq	%	Completed Year	Freq	%	Industry	Freq	%
ERP	85	53	2004	26	16	Manufacturing	67	42
KMS	14	9	2003	66	41	IT/Telecom	35	22
CRM	18	11	2002	46	28	Financial	25	15
SCM	4	2	2001	14	9	Distribution	11	7
D/W	6	4	2000	6	4	Service	15	9
EIP	3	2	1999	4	2	Others	8	5
Others	31	19	-	-	-	-	-	-
Total	161	100	Total	161	100	Total	161	100

* ERP: Enterprise resource planning; KMS: Knowledge management system; CRM: Customer relationship management; SCM: Supply chain management; D/W: Data warehousing; EIP: Enterprise information portal

Table 3. Profiles of surveyed firms

Sales	Frequency	%	# Employees	Frequency	%
>1B	45	28	> 3,000	35	22
500M-1B	21	13	1,000-3,000	35	22
300M-500M	22	14	300-1,000	27	17
100M-300M	21	13	100-300	39	24
50M-100M	28	17	50-100	25	15
10M-50M	24	15	-	-	-
Total	161	100	Total	161	100

* M: Million dollars, B: Billion dollars

luted and risky innovation (Ko et al., 2005). Our statistics also indicate that many of the IS development activities surveyed took place in the manufacturing and IT/Telecom industries. IT projects listed in Table 2 include ERP, KMS, CRM, SCM, and data-warehousing.

From the post-survey follow-up, we learned that a project team, in general, was composed of anywhere between 10 and 15 members and was partnered with one or two external consultants. The frequency distribution of sales revenue and the number of employees are relatively even throughout the categories (Table 3). In total, 109 firms had sales larger than \$100M and 97 firms employed more than 300 people.

Forty-eight percent of respondents were entry-level workers and lower-level managers, and 52% represented mid- and upper-level managers. The project team was composed of people from both IT and non-IT departments. The relatively even distribution of project participants indicates that developers had to be familiar with business processes and management issues, and naturally many at the mid- or upper management level joined the IS projects. Meanwhile, mid- and upper-level management represented only 18% of end-users. The skewed distribution of system users in terms of their organizational status re-confirms that mostly entry-level workers and lower-level managers use transactional systems to process information

Table 4. Profiles of survey respondents

Project Team Member	Frequency	%	System User	Frequency	%
Entry-level	36	22	Entry-level	73	45
Lower Mgt.	42	26	Lower Mgt.	59	37
Mid Mgt.	45	28	Mid Mgt.	25	16
Upper Mgt.	38	24	Upper Mgt.	4	2
Total	161	100	Total	161	100
Project Team Member	Frequency	%	System User	Frequency	%
Non-IT Dept.	88	55	Non-IT Dept.	97	60
IT Dept.	73	45	IT Dept.	64	40
Total	161	100	Total	161	100

and generate reports. Profiles of the survey respondents are shown in Table 4.

6.2. Validity Testing

Before the validity test of measurement models, we examined the possibility of common method bias (CMB) because this research was conducted using a self-reported survey. CMB can be curtailed by using multiple methods of data gathering, relying on longitudinal data, and taking advantage of different types of respondents (Whitman & Woszczynski, 2004). Gathering relevant data both from the developer and the user of a system, therefore, helped us in relieving the risk of CMB. However, when a single factor or one “general” factor accounts for the majority of data variation, there is a risk of CMB (Podsakoff & Organ, 1986). In our data, each of the 10 principal constructs explained about 9.5% of the total variation (12% for the first factor and 2% for the tenth factor), indicating that the effects of CMB were negligible.

Convergence Validity: Convergence validity of indicator items is generally accepted when their factor loading is greater than 0.6 and the t-value is larger than 2.0 (Bagozzi & Yi, 1988). In our study, factor loadings confirmed the statistical significance of all indicators in

their convergence (Table 5). In addition, the measurement models that explain the second-order latent variables showed their convergence with all t-values higher than 2.0 (Figure 2).

Discriminant Validity: Discriminant validity between studied variables can be tested through different approaches (Fornell & Larcker, 1981). In this study, we used the method of comparing the average variance extracted (AVE) of a construct with the squared values of correlation coefficients between constructs. The test revealed that squared values were less than AVE, confirming conceptual discrimination among the studied variables (Table 6).

Reliability: Reliability indicates internal consistency among predictor items of a studied variable (except latent variables). For the test, we used two measures of composite reliability and AVE available from LISREL. Composite reliability is conceptually similar to Cronbach’s alpha. When composite reliability is greater than 0.6, and when AVE is larger than 0.5, the internal consistency of indicators is presumed. Here, composite reliability ranged between .781 and .931, and AVE’s small-

Table 5. Test of convergent validity

Construct	Item	CSE (t-value)	Construct	Item	CSE (t-value)	Construct	Item	CSE (t-value)
Absorptive Capacity (AC)	AC1	0.789	Training (TR)	TR1	0.826	Knowl- edge Transfer (KT)	KT1	0.712
	AC2	0.839(11.163)		TR2	0.831(11.811)		KT2	0.735(10.378)
	AC3	0.762(9.962)		TR3	0.821(11.919)		KT3	0.667(8.233)
	AC4	0.776(10.278)		TR4	0.724(10.253)		KT4	0.793(8.366)
	AC5	0.788(10.463)		TR5	0.745(12.092)		KT5	0.750(8.017)
PM Lead- ership (PL)	PL1	0.770	Team Trust (TT)	TT1	0.871	System Quality (SQ)	KT6	0.663(6.190)
	PL2	0.864(13.952)		TT2	0.895(14.336)		KT7	0.729(9.115)
	PL3	0.903(12.501)		TT3	0.779(11.530)		SQ1	0.800
	PL4	0.833(11.212)		TT4	0.825(13.617)		SQ2	0.832(6.279)
	PL5	0.862(11.984)	Effect'ness of Comm'n (EC)	EC1	0.885	System Quality (SQ)	SQ3	0.460(11.339)
	PL6	0.768(10.312)		EC2	0.925(17.713)		SQ4	0.863(10.53)
	PL7	0.591(7.829)		EC3	0.884(15.702)		SQ5	0.857(12.22)
	PL8	0.740(9.583)		EC4	0.766(11.856)		UB1	0.909
Team Climate (TC)	TC1	0.807	Level of Conflict (LC)	LC1	0.812	User Benefits (UB)	UB2	0.930(19.905)
	TC2	0.694(8.631)		LC2	0.863(16.518)		UB3	0.927(19.876)
	TC3	0.610(8.497)		LC3	0.933(14.272)		UB4	0.821(17.713)
	TC4	0.710(9.535)		LC4	0.863(13.171)		UB5	0.665(9.039)
-	-	-	-	-	-		UB6	0.751(10.159)

est value was .544, confirming internal consistency among the indicator items of variables in our study (Table 7).

6.3. Analysis Results

We used one-step analysis, in which measurement and structural models are estimated simultaneously. In general, one-step analysis is applied when a research model is grounded on a theory and when there exists measurement reliability in the data set. Path coefficients are shown in Figure 2, and relevant statistics that indicate the fitness of the structural equation model are summarized as:

Chi-square(X^2): 1380.48 (df=1258), P-value: 0.085

RMSEA(<0.05): 0.025, Normed X^2 (<3): 1.097 (X^2 /df),

GFI(>.90): 0.75, AGFI(>.80): 0.73

PGFI(>.50): 0.68, SRMR(<.05): 0.067

NNFI(>.90): 0.93, CFI(>.90): 0.94

Among the fit indices, only the Chi-square test is regarded as a true statistical estimator of fitness for a structural equation model (Joreskog and Sorbom 1989). The P-value (0.085) indicates the fit of the structural model. As for other fitness indexes, GFI (0.75) and AGFI (0.73) are lower than the suggested values, but all others surpass thresholds. It may be somewhat unrealistic to expect that all values meet target ranges (Boudreau et al., 2001) and, accordingly, we conclude that the model achieved convincing goodness-of-fit.

Figure 2. Path coefficients

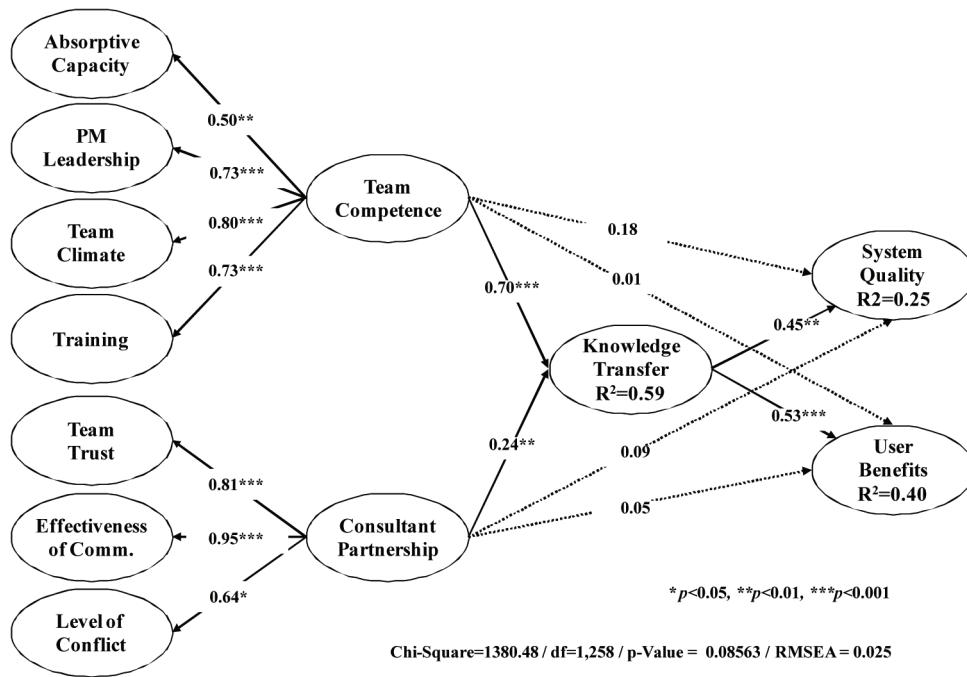


Table 6. Test of discriminant validity

	AC	PL	TC	TR	TT	EC	LC	KT	SQ	UB
AC	0.786									
PL	0.261	0.812								
TC	0.267	0.576	0.738							
TR	0.244	0.527	0.538	0.783						
TT	0.095	0.205	0.210	0.192	0.834					
EC	0.106	0.230	0.235	0.215	0.768	0.857				
LC	0.022	0.048	0.049	0.045	0.161	0.180	0.868			
KT	0.265	0.573	0.585	0.535	0.363	0.407	0.085	0.770		
SQ	0.117	0.254	0.259	0.237	0.236	0.265	0.055	0.479	0.754	
UB	0.208	0.449	0.458	0.419	0.308	0.345	0.072	0.619	0.806	0.767

* Diagonal values are rooted AVE (Average variance extracted).

* AC: Absorptive capacity; PL: PM leadership; TC: Team climate; TR: Training;

TT: Team trust; EC: Effectiveness of communication; LC: Levels of conflict; KT: Knowledge transfer; SQ: System quality; UB: User benefits

Table 7. Reliability test

Variables	Composite reliability (> 0.60)	AVE (> 0.50)
Absorptive capacity (AC)	0.890	0.619
PM leadership (PL)	0.931	0.660
Team climate (TC)	0.781	0.544
Training (TR)	0.888	0.613
Team trust (TT)	0.901	0.696
Effectiveness of comm. (EC)	0.916	0.734
Levels of conflict (LC)	0.925	0.754
Knowledge transfer (KT)	0.888	0.593
System quality (SQ)	0.790	0.568
User benefits (UB)	0.810	0.588

* AVE: Average variance extracted

The paths from team competence and consultant partnership to knowledge transfer are statistically significant at $P=0.01$. The paths from knowledge transfer to project success (system quality and user benefits) are also significant. However, none of the direct paths from team competence and consultant partnership to project success was statistically significant. The result reveals that knowledge transfer is fully mediating the links. This fact highlights that knowledge transfer from the consulting partner to the project team and the team's efforts toward knowledge internalization and institutionalization had a tremendous effect on system quality and user benefits. Also, knowledge transfer played a bigger role in project success than the other two variables did. Lastly, team competence had a greater impact on knowledge transfer and the post-implementation success of an IS project than consultant partnership did. Table 8 summarizes the results of hypothesis testing.

7. DISCUSSION

This study was undertaken to examine the effect that team competence and consultant partnerships have on intra- and inter-firm knowledge transfer and its mediating role in

shaping the post-implementation success of an IS project. The data analysis confirmed the importance of knowledge delegation to enhanced system quality and user benefits. Overall, effectiveness in knowledge transfer had a much greater impact on user perceptions of system performance than team competence and consultant partnership did. Although team competence and consultant partnership were important, they did not directly affect the success of an IS project. Rather, their influence was fully mediated by the knowledge transfer variable. R-square values indicate that 24.9% and 39.8% of variations in the system quality and user benefits variables were explained by the effectiveness of knowledge transfer.

The mediation effect is conceptually well-grounded in that team competence and consultant partnership set the tone for the intra- and inter-organizational distribution of system knowledge. In fact, their manifest variables have been frequently discussed as critical success factors of an IS project (White & Leifer, 1986). They are, however, relatively static forces compared to knowledge transfer, which is more dynamic and procedural in nature. The study result seems to reassure that the outcome of an IS project is not so much influenced by its static elements (i.e., resources and capability) as its

Table 8. Summary of hypothesis testing

Hyp.	Path			Coef.	SD	t-value	Result
H1	Team competence	→	Knowledge transfer	0.697	0.102	6.850***	Accepted
H4	Consultant partnership	→	Knowledge transfer	0.244	0.077	3.157**	Accepted
H7	Knowledge transfer	→	System quality	0.450	0.163	2.753**	Accepted
H8	Knowledge transfer	→	User benefits	0.534	0.149	3.592***	Accepted
H2	Team competence	→	System quality	0.181	0.153	0.119	Not accepted
H3	Team competence	→	User benefits	0.010	0.137	0.750	Not accepted
H5	Consultant partnership	→	System quality	0.088	0.087	1.002	Not accepted
H6	Consultant partnership	→	User benefits	0.050	0.078	0.644	Not accepted
R ² : 0.59(knowledge transfer)/ 0.25(system quality)/0.40(user benefits)							

*p<0.05, **p<0.01, ***p<0.001

procedural quality (i.e., knowledge flow and integration, development of common language and identity). By adding the knowledge transfer construct, our research supplements many extant studies that examined the normative relationship between critical success factors (CSF) and project success. Furthermore, our study confirms that the feasibility of CSF during IS implementation does not necessarily translate into its post-implementation success if the procedural integrity falls short, especially in terms of knowledge spanning across various boundaries. For practitioners, therefore, the research results suggest that a systematic plan for knowledge transfer practices (i.e., access, internalization, and institutionalization) should be in place to ensure an IS project's prolonged success.

It was found that team competence had a greater influence on knowledge transfer and therefore played a much bigger role in the ultimate success of an IS project than consultant partnership did. The R-square value of knowledge transfer, 0.59, indicates that about 59% of the variation in knowledge transfer was explained by the two constructs. To a certain degree, the influence gap between team competence and consultant partnership may be natural because the client team is responsible

not only for the initial acquisition of knowledge, but also for its translation, internal sharing and institutionalization through internal procedures and boundary objects. In contrast, consultants' principal responsibility is the delegation of system know-how to the team so that the latter can undertake an IS project. If the project team lacks dynamic and combinative capability (Mitchell, 2006) to span IS knowledge across individual, team, department, and firm-level boundaries, the effect of consultants' partnership on project success can be substantially weakened (Cohen & Levinthal, 1990). This reconfirms the fact that the partnership effect on the post-implementation success of an IS project can be marginal if a team lacks competence to carry the project through (O'Dell & Grayson, 1998).

As the manifest variables of team competence, absorptive capacity, PM leadership, team climate, and training were identified. Although all variables were statistically significant, absorptive capacity was a relatively weaker predictor of competence for knowledge transfer than the other variables. Absorptive capacity is the team's readiness to absorb domain knowledge, and is measured through questions about the respondents' prior experience and knowledge of the IS project (Cohen & Levinthal, 1990).

The analysis result (Figure 2) implies that a team's prior experience and knowledge, although important, may not effectively translate into team competence in knowledge spanning across organizational boundaries at various levels. Rather, offering adequate training to team members, having the right PM who recognizes the importance of knowledge transfer and exercises his/her leadership skills, and cultivating a team culture that promotes collaboration and knowledge sharing are more powerful in enhancing a team's competence in knowledge transfer.

As the essential conditions of consultant partnership, team trust, communication effectiveness, and conflict levels were chosen. Statistical analysis confirmed that partnership quality was reflected in a team's confidence in the consulting partner, effectiveness of communication between the team and the consultant, and healthy conflict resolution between them. Good partnerships positively influenced effectiveness in knowledge transfer and, subsequently, system success. Surprisingly, the conflict variable was a positive indicator of consultant partnership quality, although it was a much weaker predictor than the other two. This might imply that, although conflicts are more often regarded as detrimental to a partnership (Ehtamo et al., 1999), their effect may be either neutralized or turned into a positive force when managed adequately (Irish, 1975). This seemingly paradoxical development may occur when the partnership can leverage differences among partners and balance their divergent concerns for synergistic progress (Levina, 2005).

This study has several limitations. First, IS projects surveyed include ERP (enterprise resource planning), SCM (supply chain management), CRM (customer relationship management), KMS (knowledge management system), and others that may be considerably different in terms of system complexity, project duration, risks, and other contextual properties (i.e., uncertainties). Future research can achieve higher external validity by controlling variables (i.e., size and type of IS projects, firm size) that could compound statistical estimation. Meanwhile,

our research grounded on system projects and organization types that are heterogeneous in nature complements the findings of previous research undertaken in a more homogeneous setup such as ERP project (Wang et al., 2007).

The study did not seek responses from the consultants who were involved in the IS project of studied firms. Gathering three-way data from consultants, project team members, and end-users would have made the empirical study much more solid by eradicating potential common method bias (Whitman & Woszczynski, 2004). There was, however, a practical limitation in locating all three parties (developer, consultant, and user) of the relevant IS projects. In addition, studies of this kind can only be truly rigorous and reliable when the longitudinal investigation of the project life cycle becomes an integral part of the research design, rather than purely relying on snapshot data. Lastly, as always, when data are gathered from a single country, readers have to be cautious in interpreting the analysis results because of their cultural implications. The limitations outlined here reveal directions for future research.

Above all, in conducting research on knowledge transfer, our focus was on understanding relationships between selected social forces and IS success. Future research can incorporate other non-social elements (Robey & Orlikowski, 1991), such as IT infrastructure and institutional properties, to examine their implications on knowledge transfer and subsequent project performance. Also, understanding the role of particular knowledge transfer mechanisms in their composition and intensity (Slaughter & Kirsch, 2006) and their effects on the performance of an IS project can be examined in future studies. Our research can also be extended by dissecting the transfer mechanism of IS knowledge in terms of its procedural components, and understanding their relationships. For example, knowledge transfer itself may be divided into such essential elements as knowledge acquisition, transformation, internalization, institutionalization, sharing, and usage. With the process-oriented research model and its validation, we can better comprehend

the dynamic environments in which knowledge management activities trigger IS success.

8. CONCLUSION

The importance of knowledge-driven project management in affecting the outcome of an IS project has been demonstrated in a number of studies. When a project team facilitates spanning of system knowledge through boundary objects and social practices, we expect organizational 'fit' among people (i.e., end-users), the system, and user tasks to be enhanced. A team undertaking an IS project assisted by external consultants is responsible not only for its completion and delivery, but also for related subsequent post-implementation activities, including operation, maintenance, and support for business clients. One of the project team's responsibilities is to mediate system knowledge through intra- (project team and business clients) and inter- (consultants and project team) organizational boundaries via various boundary objects and routines. Knowledge transfer in this context may be both dyadic (i.e., consultants and project team) and systematic (i.e., project team and business clients). The consultant in the consultant-project team relationship, and the project team in the team-business clients relationship, become knowledge sellers or providers. Our empirical work confirmed the crucial role of team competence for knowledge transfer and consultant partnership as a joint effort; in addition, knowledge transfer itself turned out to be the most critical to the post-implementation success of an IS project.

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Sanghoon Lee is a Senior Researcher of the Korea Institute of Procurement. He earned a PhD in Business Administration (Information Systems) from Yonsei University in Korea. He was a postdoctoral research scholar at the Moore School of Business at University of South Carolina. His current research interests include e-Business, mobile service, and the information privacy/security. He has published articles in Journal of AIS, Asian Pacific Journal of Information Systems, Information Systems Review and other academic journals.

Bongsik Shin is a Professor at the Department of Information and Decision Systems at the San Diego State University. He earned a Ph.D. from the University of Arizona. He conducts research in IT-driven business models, e-commerce, and research methodology. His work has been published in respected journals including MIS Quarterly, IEEE Transactions on Engineering Management, IEEE Transactions on Systems, Man, and Cybernetics, Journal of Association for Information Systems, Communications of the ACM, Journal of Management Information Systems, Information Systems Journal, Information & Management, Decision Support Systems, and Journal of Organizational Computing and Electronic Commerce.

Ho Geun Lee is Professor of School of Business at Yonsei University in Seoul, Korea. He received his Ph.D. in MIS from the University of Texas at Austin. His research area includes inter-organizational systems, e-commerce, ubiquitous networks, and IT productivity. He is currently serving as an Editor-in-Chief of Asia Pacific Journal of Information Systems. His articles appeared in a number of professional IS journals including Information Systems Research, Communications of the ACM, Journal of Management Information Systems, Information & Management, International Journal of Electronic Commerce, Information Systems Journal, International Journal of Electronic Markets, Information Systems Frontier, Journal of Organizational Computing and Electronic Commerce, and Decision Support Systems.

APPENDIX A. SURVEY QUESTIONNAIRE

- Experience with information systems projects
 1. Have you participated in any IS project(s) at the company where you are currently working?
Yes[], No[]
 2. What is the IS project you participated in most recently at the company where you are currently working?
ERP[], CRM[], SCM[], KMS[], EIP[], other[]
 3. In what year was the project in question 2 completed?
2004[], 2003[], 2002[], 2001[], 2000[], 1999[], other[]
- Questions on team competence
 1. Absorptive capacity
 - The team members had the knowledge necessary to conduct the project.
 - The team members had experience from earlier projects.
 - The team members had participated in earlier projects that hired external consultants.
 - The team members applied project knowledge to conduct organizational duties.
 - The team members had project knowledge obtained from prior experience.
 2. PM (project manager) leadership
 - The PM clearly understood the objectives, schedule, and budget requirements of the project.
 - The PM set clear and detailed expectations.
 - The PM defined resources and inputs precisely.
 - The PM regularly provided feedback on important measures of planned and current levels of goal achievement.
 - The PM defined rules and norms for teamwork.
 - The PM actively tried to solve conflicts that occur in the team.
 - The PM was allowed to make important decisions on the project.
 - The PM understood responsibilities and competencies of the team member.
 3. Team climate
 - The team actively adopted new technologies and knowledge.
 - There was a healthy tension among members, which fostered team performance.
 - Each member's autonomy in decision-making and project activities was respected.
 - Team members collaborated to achieve project goals.
 4. Level of training
 - The team was educated on the effects of the IS project on the firm's business activities.
 - The team received training about the development and operation of the target system.
 - The team was educated about the strategic usage of the target system.
 - The team received training to improve teamwork during the IS project.
 - The team was educated about the importance of the project in addressing problems the firm was facing.
- Questions on consultant partnership
 1. Trust in the partner
 - The partner made decisions favorable to our firm.
 - The partner offered solutions for the problems the team encountered during the project.
 - The partner made efforts to conform to service level agreements.
 - The partner earned our trust in his/her competence.

2. Effectiveness in communication

Communication between the partner and the team was timely.

Communication between the partner and the team was accurate.

Communication between the partner and the team was appropriate.

The partner(s) and the team collaborated through frequent meetings.

3. Level of conflict between the partner and the team

- There were conflicts in defining task boundaries.

- There were conflicts in defining roles and responsibilities.

- There were conflicts in the manner in which tasks were undertaken.

- Overall, there were significant conflicts during the project.

Effectiveness in knowledge transfer

- Team members were used to share new ideas and methodology.

- When the team encountered operational problems, the team looked at other departments' experience to get access to critical information and knowledge.

- The team recognized the value of ideas and methods acquired from the project.

- Knowledge acquired from the project was documented and made available organization-wide.

- People shared acquired knowledge on what works or doesn't work to develop new ideas.

- Knowledge acquired from the project has been helpful to render professional services to customers.

- Efforts were made to spread acquired knowledge throughout the organization to solve organizational problems.

Project outcomes

1. System quality

Users made use of the system.

The system met intended user needs.

The system had minimal start-up problems.

Users were satisfied with the system.

Users accepted the system.

2. User benefits

The system improved task performance.

The system improved task efficiency.

The system improved task effectiveness.

The system improved decision-making.

The system improved organizational learning.

The system made desired information available.

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